INFO 4220 – Networks II: Market Design
Spring 2022 – Problem Set 2

Due date: (Friday) February 25, 11.59PM EST

Late submissions will be penalized with 1 point for every hour late.

Instructions: For each question, show all your work or you will lose some or all of the credit for that question. ***Just showing the result without showing and explaining all steps you took to find it does not count.***

How to submit: Solutions must be submitted using Gradescope. For instructions on how to submit the assignment, go to this guide or watch this video.

Important: You must select the pages for each question in Gradescope. If you don't do this, you may be penalized with 3 points. If you don't know how to do this, read this guide.

Asking questions in Ed Discussions: You can use Ed Discussions to ask clarification questions about the problems. You cannot ask on Ed Discussions how to solve a particular problem, and specially, you cannot post on Ed Discussions (or anywhere else) your solution to the problem set (or a subset of it). Disclosing the solution of problems in Ed Discussions (or anywhere else) will be considered an academic integrity violation and reported accordingly. If you don't know how to solve a problem, don't ask on Ed Discussions, come to office hours! (Although, we will not solve the problem for you)

Who did you work with to solve this PS, if anyone (remember each of you must turn in their own solution, which has to be your own original work)? (0 pt)

Are you using and slip day for this PS? Remember that you have 2 for the entire semester, and they can only be used in 1-day increments. (0 pt)

Problem 1 (20 points)

Consider a medical match problem with three hospitals, h_1, h_2, and h_3, and five doctors, d_1, d_2, d_3, d_4 and d_5. Hospital 1 is in a rural area can admit up to 1 doctor, Hospital 2 in a suburban region can admit up to 2 doctors, and Hospital 3, in a larger city, can admit up to 3 doctors. The preferences of doctors over hospitals and hospitals over doctors are given by the following tables:

$\boldsymbol{d_1}$	$\boldsymbol{d_2}$	$\boldsymbol{d_3}$	$\boldsymbol{d_4}$	$\boldsymbol{d_5}$
h_3	h_3	h_3	h_3	h_2
h_1	h_2	h_1	h_1	h_1
h_2	h_1	h_2	h_2	h_3

$\boldsymbol{h_1}$	$\boldsymbol{h_2}$	$\boldsymbol{h_3}$
d_4	d_5	d_5
d_2	d_3	d_1
d_1	d_1	d_2
d_3	d_2	d_3
d_5	d_4	d_4

a) (6 points) Find the matching that would result from using the deferred acceptance algorithm with doctors proposing.

Step 1: $d_1 \to h_3, d_2 \to h_3, d_3 \to h_3, d_4 \to h_3, d_5 \to h_2$. *Rejections:* h_3 *rejects* d_4.

Step 2: $d_4 \to h_1$. *There are no rejections, so the final matching is* $d_1 \to h_3, d_2 \to h_3, d_3 \to h_3, d_4 \to h_1, d_5 \to h_2$ *or* $h_1 \to \{d_4\}, h_2 \to \{d_5\}, h_3 \to \{d_1, d_2, d_3\}$.

+ 3 points for the final matching

+ 3 points for the correct process with 2 steps

b) (6 points) The hospitals are not happy with the outcomes that the algorithm produces. Over the years there is always a hospital that ends up with an empty spot, and it is often hospitals 1 or 2 that have a great need for doctors. One time, during several years in a row, hospital 1 couldn't recruit any doctors and was severely understaffed. The director of the medical association, which is an NFL fan, proposes to take a hint from the NFL draft mechanism and proposes the following: Hospitals will be matched to doctors one at a time based on the preferences they submitted and according to the following order: The hospital that did not fill its vacancies the previous year will be matched to their most preferred doctor first. The order in which the other hospitals will be matched to doctors is based on a lottery. If after going over all hospitals there are still doctors unmatched, a new round of matchings starts following the same order. The algorithm ends when there are no more doctors to assign. Last year hospital 1 did not fill its vacancies, and the lottery determined that hospital 3 will pick ahead of hospital 2 (i.e. hospitals will pick doctors in this order: h_1, h_3, h_2). What will be the resulting matching if hospitals submit their true preferences (i.e. when it is their turn, they pick the available doctors that is highest in their list)

In the first round: $h_1 \to d_4$, $h_3 \to d_5$, and $h_2 \to d_3$

In the second round, hospital 1 is already full. Only hospitals 2 and 3 make offers: $h_3 \to d_1$, $h_2 \to d_2$

In the last round only hospital 3 has a vacancy, but there are no more doctors left.

The final matching is $\mu(h_1) = d_4, \mu(h_2) = \{d_3, d_2\}, \mu(h_3) = \{d_1, d_5\}$

+ 3 points for the final matching

+ 3 points for the correct process with 2 steps

c) (4 points) Is the matching you found in (b) stable? Demonstrate why or why not.

Not, it is not stable because d_4 prefers h_3 to h_1, and h_3 still has a vacancy. This means that the matching is wasteful.

+ 2 the resulting matching is not stable

+ 2 the resulting matching is not stable because d_4 prefers h_3 to h_1, and h_3 still has a vacancy. This means that the matching is wasteful. Note that other blocking pairs could be accepted.

d) (4 points) Is the mechanism proposed in (b) strategy-proof. Demonstrate why or why not.

No, it is not strategy-proof. Hospital 2 would benefit from ranking d_1 ahead of d_3. In that way, in the first round they pick d_1 before hospital 3 gets a chance to hire them in the next round, and they don't lose anything because they can still get d_3 in the following round.

+ 2 the resulting matching is not strategy-proof

+ 2 explain the reason why the matching is not strategy-proof

Problem 2 (20 points)

College Town Communities (CTC) is one of the largest (fictional) off-campus student apartments in Ithaca. Many Cornell students live there due to the well-equipped facilities and convenient shuttle bus to school. Usually starting from March every year, students start to consider where to live next academic year and start to reserve apartments. Let's say there are 4 Cornell students, i_1, i_2, i_3 and i_4, already living in College Town Communities this semester with houses h_1, h_2, h_3, h_4. They are considering renewing the current apartment or moving to another apartment at CTC. At the same time, a newly admitted student at Cornell, i_5 , is also considering reserving an apartment at CTC. Nancy, the leasing agent at the CTC apartment management office, says that for the next academic year h_1, h_2, h_3, h_4 will be made available for reservation if the current tenants choose not to renew, and that houses h5, h6, h7 are currently vacant and available for reservation. Let's help Nancy design a solution to allocate the

houses to the 5 students. Assume the five students' preference over the 7 apartment is as follows:

$\mathbf{i_1}$	$\boldsymbol{i_2}$	$\mathbf{i_3}$	$\mathbf{i_4}$	$\boldsymbol{i_5}$
h2	h7	h2	h2	h4
h6	h1	h1	h4	h3
h5	h6	h4	h3	h7
h1	h5	h7	h6	h1
h4	h4	h3	h1	h2
h3	h3	h6	h7	h5
h7	h2	h5	h5	h6

a) (6 points) The first way Nancy raises up is to draw from 5 cards with 5 different numbers written. Each of the students picks one card. Whoever picks the largest number picks apartment first. Then the person that gets the second largest number chooses second among the remaining rooms and so on in a similar fashion. Tenants i_1, i_2, i_3, i_4 are hesitant whether they should just keep their current houses or participate in this random serial dictator allocation mechanism. Assume that if i_1, i_2, i_3, i_4 participate, then the dictators' order will be i_1, i_2, i_5, i_3, i_4. What is your suggestion to i_1, i_2, i_3, i_4 on whether they should participate in the random serial dictator mechanism or keep their houses?

The assignment would be as follows: $\mu(i_1) = h_2, \mu(i_2) = h_7, \mu(i_3) = h_1, \mu(i_4) = h_3, \mu(i_5) = h_4$

Tenant i_4 *ends up with an assignment that they like less than their initial hose, so the assignment is not individually rational and* i_4 *should not choose to participate if that was the dictators' ordering.*

+ 3 for the correct assignment

+ 3 for the recommendation to i_4 *to not participate in the matching*

b) (4 points) Is the assignment in part (a) efficient? Why or why not?

Yes, it is efficient because you cannot make anyone better off without making anyone else worse off.

+ 2 the assignment is efficient

+ 2 for the justification: with the assignment, you cannot make anyone better off without making anyone else worse off

c) (6 points) Another way Nancy raises up to allocate the apartments is to use the YRMH-IGYT algorithm. Let's say the dictators' order is i1, i2, i5, i3, i4. What is the resulting assignment? Show your work step by step.

At first, i1 chooses h2, which conflicts with the tenant i2.
According to YRMH-IGYT algorithm, we assign a new order: i2, i1, i5, i3, i4.
Then, i2 chooses h7, i1 chooses h2, i5 chooses h4 which conflicts with i4.
We cancel the assignments from i4 and reorder the remaining tenants i4, i5, i3.
Then, i4 chooses h4, i5 chooses h3 which conflicts with i3.
We cancel the last assignment and reorder tenants as: i3, i5.
Then, i3 chooses h1, i5 chooses h3.
Final assignment: $\boldsymbol{i_1 \rightarrow h_2, i_2 \rightarrow h_7, i_3 \rightarrow h_1, i_4 \rightarrow h_4, i_5 \rightarrow h_3}$.

+ 3 for the final assignment

+ 3 for the correct process with all steps listed

d) (4 points) Is the assignment you found in (c) the best possible assignment that is both efficient and individually rational for tenant i_4? Can tenant i_4 lie about their preferences in order to obtain a better house?

No, it is not the best possible assignment that is both efficient and individually rational for tenant i_4. *If the random draw resulted in the ordering:* i_4, i_3, i_5, i_2, i_1 *the resulting assignment would be different*.

i_4 *will choose* h_2, *which will create a conflict with* i_2. *Thus assignments are cancelled and we re-order the dictators:* i_2, i_4, i_3, i_5, i_1.
Now i_2 *chooses* h_7, i_4 *chooses* h_2, *and* i_3 *will choose* h_1, *which creates a conflict with* i_1.
We thus cancel the assignments from i_3 *and re-order the remaining dictators as:* i_1, i_3, i_5
We resume the serial dictatorship and i_1 *will choose* h_6, i_3 *will choose* h_1 *and finally* i_5 *chooses* h_4.
Final assignment is: $\boldsymbol{i_1 \rightarrow h_6, i_2 \rightarrow h_7, i_3 \rightarrow h_1, i_4 \rightarrow h_2, i_5 \rightarrow h_4}$

Note that the only way for i_4 *to obtain this better assignment is by "luck" in the random draw that determines the order in which houses are allocated.*

+ 1 the resulting assignment is not the best possible assignment that is both efficient and individually rational for tenant i_4

+ 1 finds an instance or "random" draw where i_4 *is better off; must show the order and steps to get full credit*

+2 states that i_4 *cannot lie about their preferences to obtain a better house*

Problem 3 (10 points)

An unnamed university in a rural town near the border with Canada is considering the following system to allocate dorm rooms. Each student must submit a preference list over which buildings they would like to live at the next year. Let's assume there are 5 buildings (North, South, East, West, and Central) and 5 students: Antonio (freshman), Bruno(senior), Camilo (sophomore), Dolores (senior), and Isabela (junior). Assume each building has a capacity of one. Assignments are determined using the top trading cycle algorithm considering that all rooms are public endowment and that students get different priorities over buildings according to their seniority (and if two students have the same seniority, their priority is determined by alphabetical ordering). At North and South freshmen get the first priority, then sophomores, then juniors, and finally seniors. At East and West Seniors get the first priority, then juniors, then sophomores, and lastly freshmen. At Central, juniors get first priority, then sophomores, then seniors, and finally freshmen. The students submitted the following preferences:

Antonio	***Bruno***	**Camilo**	**Dolores**	***Isabela***
South	North	South	South	West
Central	Central	North	West	East
North	West	West	East	North
West	East	East	North	South
East	South	Central	Central	Central

a) (10 points) Find the assignment according to the mechanism described above:

The first thing we need to do is to construct a table with the priorities at each building

North	***South***	***East***	***West***	***Central***
Antonio	*Antonio*	*Bruno*	*Bruno*	*Isabela*
Camilo	*Camilo*	*Dolores*	*Dolores*	*Camilo*
Isabela	*Isabela*	*Isabela*	*Isabela*	*Bruno*
Bruno	*Bruno*	*Camilo*	*Camilo*	*Dolores*
Dolores	*Dolores*	*Antonio*	*Antonio*	*Antonio*

We now run the TTC algorithm for problems with objects with priorities:

In the first step: Antonio is assigned to South

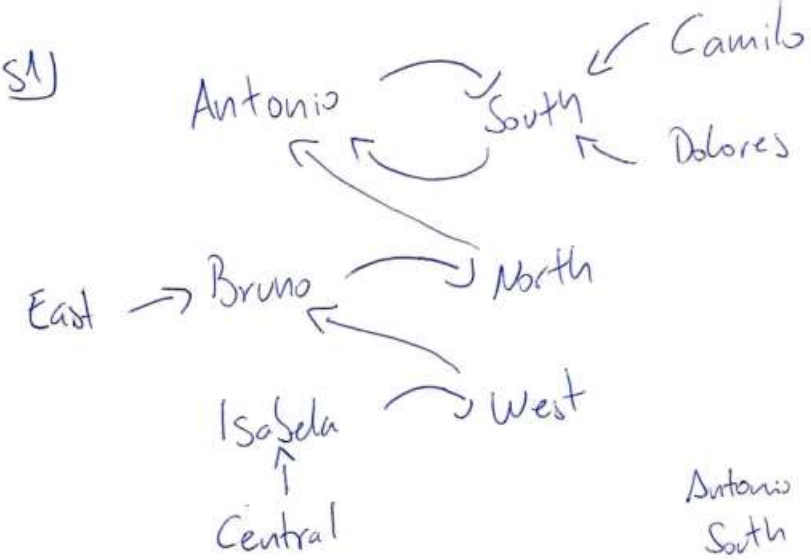
In the second step: Camilo is assigned to North

In the third step: Bruno is assigned to Central, and Isabela to West

In the fourth step: Dolores is assigned to East

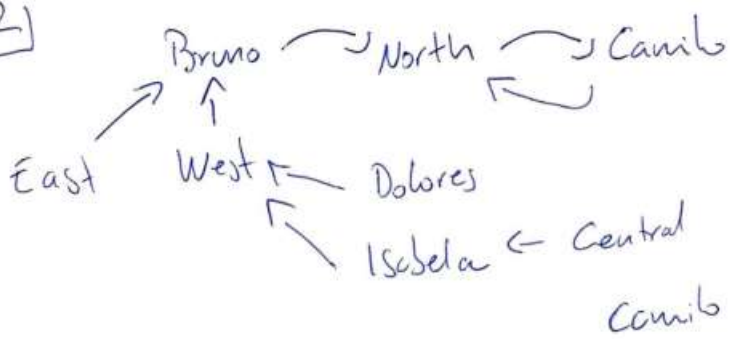
+ 2.5 for each step along with the corresponding diagram

S1)



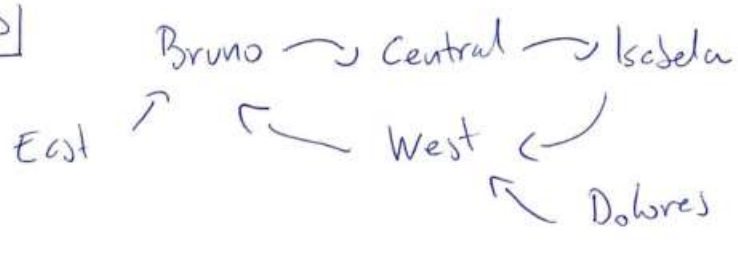
Antonio assigned to South

S2)



Camilo assigned to North

S3)



Bruno assigned to Central

Isabela assigned to West

S4)



Dolores assigned to East